

BINAY - Lighting Up The Future



BINAY LED-based HIGH, MEDIUM and LOW Intensity Aviation Obstruction Light Beacons

As per International Civil Aviation Organisation (ICAO) requirements Available in Low Intensity, Medium Intensity and High Intensity versions (as per International Civil Aviation Organization guidelines), BINAY's patented LED Aviation Lights come with 5-year/3-year warranties

LED Obstruction Lighting for:

- Industrial chimneys and smokestacks
- Transmission, microwave and cellular towers
- Radio, TV and similar structural towers
- High-rise buildings and structures
- Airports and airfields

The BINAY LED Aviation Obstruction Light offers the following advantages:

- Fit-and-forget maintenance-free operation
- A long life of 100,000 hours (20 years at 12 hours daily burning)
- Pays for itself within a short period of operation in the form of reduced installation, maintenance and servicing costs
- Quick Installation; Reliable operation 365 days per year
- Shock-proof and vibration-resistant
- Over-Designed Intensity to allow for natural LED Intensity degradation over its operating lifetime



THE BINAY LED OBSTRUCTION LIGHT IS UNDER ACCEPTED PATENT, AND AS SUCH IS A PROPRIETARY PRODUCT

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POWERING LED TECHNOLOGIES WORLDWIDE SINCE 1983

To develop the system, Ashutosh Saxena, assistant professor of computer science at Cornell University, and colleagues recorded videos of ten drivers, along with videos of the road ahead, for 1899km of expressway and city-driving over a period of two months.

A computer using face-detection and tracking-software identified head movements and learnt to associate these with turns and lane changes, so that the final system can anticipate possible actions the driver may take.

Acoustruments use sound waves to control smartphones

Researchers have developed an inexpensive alternative to smartphone touchscreens—a toolbox of physical knobs, sliders and other mechanisms that can be readily added to control any device.

Researchers at Carnegie Mellon University and Disney Research drew inspiration from wind instruments in devising these mechanisms, which they call acoustruments.

Their idea is to use pluggable plastic tubes and other structures to connect a smartphone's speaker with its microphone. The device can then be controlled by acoustically altering sounds as these pass through this system. Just as a simple slide whistle or flute can produce expressive music, these acoustruments can add a wide range of functionality to a smartphone including proximity and pressure sensors.

The plastic acoustruments can be made rapidly and inexpensively as no electrical circuitry is involved.

Engineers create world's smallest beamsplitter

As computer technology is forging ahead, demand for processing information at ultra-fast speed is also increasing. Scientists believe that silicon photonics has the potential of improving performance and speed of machines such as supercomputers. Instead of using electrons for transferring and manipulating information, this technology uses photons of laser light. Engineers from University of Utah have taken a step in the direction of developing computers and mobile devices that could compute information at the speed of light, that is, millions of times faster than present-day machines.

They have designed an integrated-nanophotonics polarisation beamsplitter on top of a silicon chip that can divide incoming light waves in two. As a result, two distinct channels of separately polarised information are generated. The extremely-small beamsplitter is the world's smallest polarisation beamsplitter, and is just 2.4 microns by 2.4 microns (approx. one-fiftieth the width of a human hair). By applying the new nonlinear optimisation algorithm, researchers were able to scale